

Indian Institute of Technology Palakkad

Department of Mathematics

MA 5015 (Numerical analysis with programming)

2026 (January-May Semester)

Instructor: Dr Gopikrishnan C R

Office: 10, D03-FF (Dr. APJ Abdul Kalam Block).

Syllabus

- ▶ *Review of elementary calculus* Mean value theorem, Rolle's theorem, Taylor's theorem with different error formulae, $C^k[a, b]$ functions.
- ▶ *Introduction to finite precision arithmetic.* Representation of numbers of finite digit machines, rounding and chopping errors, error propagation, asymptotic convergence, big and little oh notation.
- ▶ *Root finding of nonlinear systems.* Bisection, Secant and Newton's method, fixed point iteration
- ▶ *Linear systems.* Gauss elimination, LU, QR, matrix norms, Jacobi and Gauss-Seidel method
- ▶ *Interpolation.* Lagrange interpolation, Newton interpolation using divided differences, Hermite interpolation, piecewise linear and cubic spline interpolation
- ▶ *Numerical differentiation.* Finite differences, Richardson extrapolation
- ▶ *Numerical integration.* Trapezoidal rule, Simpson's rule, Gaussian quadrature in 1D and 2D
- ▶ *Numerical solutions of ordinary differential equations.* Euler's method, implicit method.

References

- [1] Robert Johansson. *Numerical Python*. Apress, [Berkeley], CA, second edition, 2019. ISBN 978-1-4842-4245-2; 978-1-4842-4246-9. doi: 10.1007/978-1-4842-4246-9. URL <https://doi.org/10.1007/978-1-4842-4246-9>. Scientific computing and data science applications with Numpy, SciPy and Matplotlib.
- [2] James F. Epperson. *An introduction to numerical methods and analysis*. John Wiley & Sons, Inc., Hoboken, NJ, third edition, 2021. ISBN 978-1-119-60469-3. With a foreword by Jean Mawhin.

Additional reading material

1. Practical numerical computing using Python, Scientific and Engineering Applications, Mahendra Verma, ISBN: 978-93-91374-51-8